RESEARCH ARTICLE

European Journal of Biology and Biotechnology
www.ejbio.org

Effects of *Ilex paraguariensis* A. St.-Hil. (Yerba Mate) and Chlorogenic Acid on Creatine Phosphokinase, Lactate and Irisin Levels in Mice after Detraining

Evandro Toledo Gerhardt Stutz, Roberto Laureano-Melo, Vladimir Lopes de Souza, Geraldo Ceni Coelho, Claudia Cardoso Netto, and Carlos Alberto Bastos de Maria

ABSTRACT

This is the first report that deals with the effect of yerba mate (YM) and chlorogenic acid (CGA) consumption on blood total creatine phosphokinase (CPK), lactate and irisin levels after a detraining period. Healthy mice (n = 50) were randomly separated into the following experimental groups: SED = sedentary control; TRAIN = mice submitted to training (swimming) for 8 weeks; DETRAIN = mice submitted to training for four weeks and after submitted to detraining for four weeks; DETRAIN CGA = mice submitted to training for four weeks and after submitted to both detraining and standard CGA consumption for four weeks; DETRAIN YM = mice submitted to training for four weeks and after submitted to both detraining and consumption of roasted YM infusion for four weeks. No significant (p > 0.05) difference was found in relation to the lactate value. The DETRAIN CGA group had a lower CPK data when compared with the other groups. The DETRAIN YM group showed a lactate value slightly lower than that from DETRAIN CGA one. Blood irisin value had no statistical (p > 0.05) difference between studied groups. Irisin value in sedentary group had a value non-statistically (p > 0.05) significant when compared with the other groups after a detraining period. Therefore, CGA could decrease the protein catabolism via CPK inhibition during detraining. YM may maintain a part of the lipid mobilization during detraining, leading to the sparing of a fraction of the lactate.

Keywords: CPK, detraining, lactate, irisin, mice, *Ylex paraguariensis.*

Published Online: January 24, 2023

ISSN: 2684-5199

DOI : 10.24018/ejbio.2023.4.1.419

E. T. G. Stutz*
Barra Mansa University Center, Brazil.
(e-mail: evandro.tgs@gmail.com)
R. Laureano-Melo
Barra Mansa University Center, Brazil.
(e-mail: laureanomelor@gmail.com)
V. L. de Souza
Barra Mansa University Center, Brazil.
(e-mail: vladilopes@gmail.com)
G. C. Coelho
Federal University of Fronteira Sul (UFFS), Brazil.
(e-mail: cenicoelho@gmail.com)
C. Cardoso Netto
Biomedical Institute, UNIRIO,Brazil.
(e-mail: claudia.c.netto@unirio.br)
C. A. B. de Maria
Biomedical Institute, UNIRIO, Brazil.
(e-mail: carreb@uol.com.br)

Corresponding Author

I. Introduction

Numerous physiological effects have been attributed to yerba mate (YM) (Junior & Morand, 2016; Riachi & de Maria, 2017). A part of the studies with YM has focused on energetic metabolism in different exercise conditions. In this context no difference was found not only for blood total CPK and lactate in different steps of light and moderate exercises but also for blood total CPK after acute eccentric bout (Alkhatib, 2014; Cazal, 2019; Moura *et al.,* 2020). Our group have not found differences in blood total CPK and irisin in mice submitted to chronic swimming. On the other hand, there was a significantly (p > 0.05) lower value of lactate to swimming mice treated with YM (de Souza *et al.,* 2021). Although there were studies associating YM with training, to our knowledge, there are no studies associating YM with detraining.

Presence of large amounts of caffeic acid conjugates, mainly chlorogenic acid (CGA), would be one of those responsible for the physiological actions of the plant. For examples, the CGA regulates negatively the glucose-6-phosphate translocase decreasing hepatic glucose release, as well as improves glucose uptake in skeletal muscle (Hernmele *et al.,* 1997; Ong *et al.,* 2013). Accelerated lipid oxidation is activated by CGA via AMPKα signaling route (Xu *et al.,* 2019).

Due to the lack of studies addressing the physiological effects of the YM and CGA under detraining conditions, our group decided to investigate the effects of one or the other on blood total CPK, lactate and irisin levels in mice after detraining.

II. Materials and Methods

A. Reagents

CPK was from "EBRAM Produtos Laboratoriais Ltda" (São Paulo, Brazil). CGA (5-caffeoylquinic acid) and all other reagents and solvents were from Sigma-Aldrich Chemical Co. (St. Louis, MO, USA).

B. Plant and CGA Preparation

Both Claudio Kovaleski and Silvio Kovaleski producers from the "Ervateira Chiru" (Chapecó, SC, Brazil) provided YM leaves. Dried and roasted YM was roasted at 180 ºC for 10 min as previously reported (Riachi *et al.,* 2018). More details with respect to the plant can be found in elsewhere (de

Souza *et al.,* 2021). Roasted YM infusion was prepared as follows: Boiling distilled water (500 mL) was added to the plant (8.751 g), mixed, capped and filtered under gravity after 15 min based on previous study (Riachi *et al.,* 2019). The concentration de YM infusion to be administrated to the mouse was as follows: Take into account the average weight of both a human being and a mouse, 70 Kg and 46 g, respectively, a final volume of 0.3285 mL was obtained to gavage administration. Concentration of 26.285 mg mL^{-1} was adjusted to a volume of 0.1 mL to reduce the stress of the mice during gavage.

The content of CGA (0.896 mg mL^{-1} YM infusion) in the YM used in the present study was described previously elsewhere (Riachi *et al.,* 2018). An aqueous solution of CGA standard was prepared and 0.1 mL of this solution was administrated via gavage. This volume contained approximately 0.0716 mg of standard CGA which paralleled with CGA value found in the volume of YM administrated to mice.

III. Animal Experimental Design

Sample size (n=50) represented by Swiss Webster male mice (*Mus musculus)* with 60 days old was established in accordance with recommendation from the Animal Experimental Ethics Committee of the "Barra Mansa University Center". More details on light: dark cycle, fed, swimming protocol, among others, are reported elsewhere (de Souza *et al.,* 2021). Animals were randomly distributed in five groups (n=10 per group) as follows: SED=sedentary mice without training. After four weeks, animals were treated with saline solution at 0.9 %; TRAIN=mice submitted to training (swimming) for eight weeks. After four weeks, animals were treated with saline solution at 0.9 %; DETRAIN=mice submitted to training (swimming) for four weeks followed by four weeks of detraining. In the period of detraining, mice were treated with saline solution at 0.9 %; DETRAIN CGA=it was similar to DETRAIN group except for substitution of saline solution for standard CGA solution during detraining period; DETRAIN YM=it was similar to DETRAIN group except for substitution of saline solution for YM infusion during detraining period.

The animals were euthanized by exsanguinations cardiac punction under anesthesia as described in previous study (de Souza *et al.,* 2021). The protocol was approved by Animal Experimental Ethics Committee of the "Barra Mansa University Center" (Barra Mansa, RJ, Brazil) under number 001/2022.

IV. CPK, Lactate, Irisin and Statistical Analyses

All the biochemical parameters were analyzed as previously reported (de Souza *et al.,* 2021). Kolmogorov-Smirnov and Shapiro-Wilk normality tests were applied. It was used ANOVA and Newman-Keuls multiple comparison test for variables with normal distribution and ANOVA or Kruskal Wallis test for variables with non-normal distribution. The level of significance was set at $p < 0.05$. The GraphPad version 5.0 (Los Altos, CA, USA) was used for statistical treatment.

V. Results and Discussion

To our knowledge, this is the first study dealing with the effect of YM on blood total CPK, lactate and irisin levels after a detraining period. In the Fig. 1 is shown the blood total CPK data in the groups studied. In general, there were not statistical ($p>0.05$) differences between treated (DETRAIN CGA and DETRAIN YM) and non-treated (SED, TRAIN and DETRAIN) groups. However, CGA treated group (DETRAIN CGA) had lower value compared to the other groups. It should be taken into account that presence of other bioactive molecules could cause antagonistic action or even that the complex matrix of YM could interfere with the biodisponibility of CGA. Considering that CGA would be capable of inhibiting CPK activity, the lower disponibility of CGA in the YM could increase enzymatic performance. The CPK inhibition caused by CGA has been speculated in volunteers with traumatic brain injury (Ribeiro *et al.,* 2017). Ong *et al.* (2013) and Fernandes *et al.* (2012) reported that CGA may be responsible by modulating both adenosine monophosphate-dependent protein kinase and paraoxonase-2 levels. The CGA appears to regulate enzymes involved in both primary and secondary metabolism.

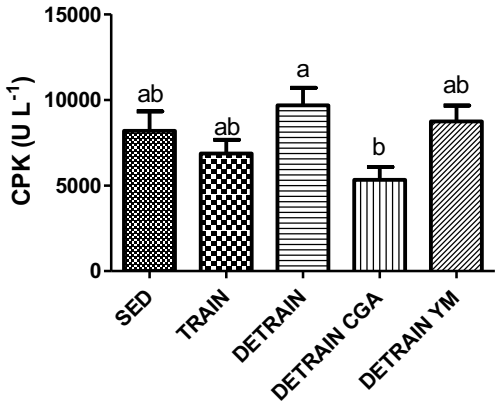

Fig. 1. Blood total CPK level in YM and CGA -treated and non-treated groups.
Abbreviations: CPK: creatine phosphokinase; YM: yerba mate; CGA: chlorogenic acid; SED: sedentary; TRAIN: training; DETRAIN: detraining; DETRAIN CGA: detraining treated with chlorogenic acid; DETRAIN YM: detraining treated with yerba mate. Different letters indicate significant differences ($p < 0.05$) between the groups. These values showed normal distribution to Kolmogorov-Smirnov and Shapiro-Wilk normality tests. ANOVA and Newman-Keuls tests were applied.

Lactate value for each mice group is presented in Fig. 2. There were not significant ($p>0.05$) differences in the treated (DETRAIN CGA and DETRAIN YM) and non-treated (SED, TRAIN AND DETRAIN) after a detraining period. Mice group treated with YM (DETRAIN YM) presented a lactate content slightly lower than that from treated with CGA (DETRAIN CGA). Presence of other phytochemicals as caffeine could act synergistically with the phenolic acid. Xu *et al.* (2019) indicated that combination of caffeine and CGA accelerated lipid oxidation via AMPKα-LXRα/SREBP-1C pathway. de Souze *et al.* (2021) showed that mice submitted to both chronic exercise and YM consumption had a significant ($p < 0.05$) lower lactate value compared to control groups. There was accelerated mobilization of lipids which could spare glycogen degradation.

Fig. 3 presents irisin values in treated (DETRAIN CGA and DETRAIN YM) and non-treated (SED, TRAIN AND

DETRAIN) groups. Results were not significatively (p>0.05) different between studied groups. In our previous study, YM-treated group and submitted to training as well sedentary and training groups showed values non-statistically (p>0.05) different from each other (de Souza *et al.,* 2021).

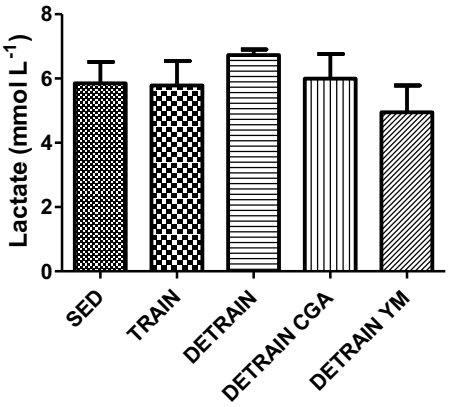

Fig. 2. Blood lactate profile in YM and CGA -treated and non-treated groups.
Abbreviations: YM: yerba mate; CGA: chlorogenic acid; SED: sedentary; TRAIN: training; DETRAIN: detraining; DETRAIN CGA: detraining treated with chlorogenic acid; DETRAIN YM: detraining treated with yerba mate. No letters indicate no significant differences (p > 0.05) between the groups. These values showed non-normal distribution to Kolmogorov-Smirnov and Shapiro-Wilk normality tests. ANOVA and Kruskal Wallis tests were applied.

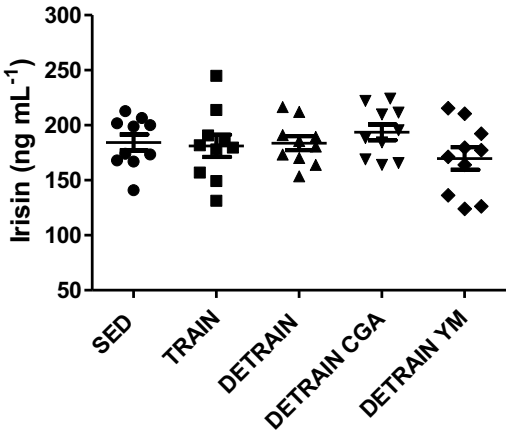

Fig. 3. Blood irisin level in YM and CGA -treated and non-treated groups.
Abbreviations: YM: yerba mate; CGA: chlorogenic acid; SED: sedentary; TRAIN: training; DETRAIN: detraining; DETRAIN CGA: detraining treated with chlorogenic acid; DETRAIN YM: detraining treated with yerba mate. No letters indicate no significant differences (p > 0.05) between the groups. These values showed normal distribution to Kolmogorov-Smirnov and Shapiro-Wilk normality tests. ANOVA and Newman-Keuls tests were applied.

In conclusion, CGA could reduce the skeletal muscle protein catabolism via CPK inhibition in the detraining phase in the experimental model used in this study. Even with the advent of detraining, YM may maintain a part of the lipid mobilization stimulated during training, leading to the sparing of a fraction of the cell's lactate. The two above-mentioned findings need further investigation. After a detraining period, irisin value in all the previously trained groups had values non-statistically (p>0.05) different compared to the sedentary group.

Conflict of Interest

Authors declare that they do not have any conflict of interest.

References

Alkhatib A. (2014). Yerba Maté (*Ilex paraguariensis)* ingestion augments fat oxidation and energy expenditure during exercise at various submaximal intensities. *Nutrition and Metabolism,* 11, 42.

Cazal, M.D.M. (2019). *Efeito da ingestão de chá mate (Ilex paraguariensis) antes do exercício aeróbico sobre indicadores metabólicos de dano muscular e estresse oxidativo* [Doctoral dissertation, Federal University of Viçosa]. Portuguese.

de Souza, V.L., Stutz, E.T.G., Pehrson, M.E., Coelho, G.C., Netto, C.C., de Maria, C.A.B. (2021). *Ilex paraguariensis* A. St.-Hil. (Yerba Mate) differently regulates the lipid mobilization and irisin and lactate levels in sedentary and chronic swimming mice. *European Journal of Medical Health Sciences,* 3(6), 58-62.

Fernandes, E.S., Machado, M.O., Becker, A.M., Andrade, F., Maraschin, M., da Silva, E.L. (2012). Yerba mate (*Ilex paraguariensis*) enhances the gene modulation and activity of paraoxonase -2: In vitro and in vivo studies. *Nutrition*, 28, 1157-1164.

Hermmele, H., Burger, H.J., Below, P., Schubert, G., Rippel, R., Schindler, P.W., *et al.* (1997). Chlorogenic acid and synthetic chlorogenic acid derivatives: novel inhibitors of hepatic glucose-6-phosphate translocase. *Journal of Medicinal Chemistry,* 40, 137-145.

Junior, E.L.C., & Morand, C. (2016). Interest of mate (*Ilex paraguariensis* A. St.- Hil.) as a new natural functional food to preserve human cardiovascular Healthy – a review. *Journal of Functional Foods*, 21, 440-444.

Moura, B.M., Panza, V.P., Brunetta, H.S., Tamborindeguy, A.C., de Oliveira M.V., Sakugawaet R.F. *et al.* (2020). Effect of mate tea consumption on rapid force production after eccentric exercise: a randomized, controlled, crossover study. *Sport Sciences for Health*, 6, 1-11.

Ong, K.W., Hsu, A., Tan, B.K.H. (2013). Anti-diabetic and anti-lipidemic effects of chlorogenic acid are mediated by AMPK activation. *Biochemical Pharmacology,* 85, 1341-1351.

Riachi, L.G., & de Maria, C.A.B. (2017). Yerba mate: An overview of physiological effects in humans. *Journal of Functional Foods,* 38, 308-320.

Riachi, L.G., Marcellini, P.S., de Maria, C.A.B., Coelho, G.C. (2019). Yerba mate consumption effect on the total concentration of creatine phosphokinase in healthy volunteers from of age of 50. *European Journal of Medical Health Sciences,* 1, 1-3.

Riachi, L.G., Simas D.R.L., Coelho, G.C., Marcellini, P.S., da Silva, A.J.R., de Maria, C.A.B. (2018). Effect of light intensity and processing conditions on bioactive compounds in mate extracted from yerba mate (*Ilex paraguariensis* A. St.-Hil.). *Food Chemistry,* 266, 317-322.

Ribeiro, M.C., Santos, A., Riachi, L.G., Rodrigues, A.C.B., Coelho, G.C., Marcellini, P.S., *et al.* (2017). The effects of roasted yerba mate *(Ilex paraguariensis* A. St. Hil.) consumption on glycemia and total serum creatine phosphokinase in patients with traumatic brain injury. *Journal of Functional Foods,* 28, 240-245.

Xu, M., Yang, L., Liao, M., Chu, L., Li, X. (2019). Collaborative effects of chlorogenic acid and caffeine in lipid metabolism via of AMPKα-LXRα/SREBP-1c pathway in high-fat diet-induced obese mice. *Food Func*tion,10, 7489-7497.

E. T. G. Stutz is Veterinary Doctor Student, Master of Science and specialist in animal morphophysiology, with experience in academic laboratories and vivarium. Personnel training, preparation of training courses, production of manuals (institutional teaching material) and standardization of procedures (POPs) in vivarium. Professor for 40 hours of the Veterinary Medicine Course at Centro Universitário de Barra Mansa -UBM, with experience in the disciplines of Bioterism, Cell Biology, Embryology, Histology and Biophysics. Technical Manager for the Animal Breeding and Experimentation Animal Facility at UBM.

R. Laureano-Melo is graduated in Veterinary Medicine from the Federal Rural University of Rio de Janeiro (2010). He holds a Master's degree and Doctorate in Physiology from the Multicenter Postgraduate Program in Physiological Sciences (UFRRJ/SBFis). He is currently an assistant professor of the Veterinary Medicine Course at the Centro Universitário de Barra Mansa (UBM) and a member of the Brazilian Society of Physiology (SBFis).

V. L. de Souza is Graduated in Physiotherapy from the Catholic University of Petrópolis (1999), Specialization in Sports and Cardiovascular Physiotherapy (2001), Specialist in Health Education for SUS Preceptors (2014), Master in Science of Human Motricity from Castelo Branco University (2006) and Doctoral Student in Nursing and Biosciences at the Federal University of Rio de Janeiro (UNIRIO) (2019). Coordinator of the Physiotherapy Course (University Center of Barra Mansa/2011). Coordinator of the Postgraduate Course in Geriatrics and Gerontology (UBM). Member of the Ethics Committee of UBM, Associate Member of the Brazilian Association of Teaching in Physiotherapy (ABENFISIO).

G. C. Coelho is Biologist, Master in Botany (UFRGS – Rio Grande do Sul Federal University, Porto Alegre, Brazil, 1996), Doctor in Pharmaceutical Sciences (UFRGS, 2002), He works on chemical ecology, ecophysiology and forest ecology. Member of the Post-Graduation in Environmental Science and Technology, Fronteira Sul Federal University (UFFS) at Erechim, Brazil.

C. Cardoso Netto is Graduated in Nutrition from the Federal University of the State of Rio de Janeiro (UNIRIO), Master's and Doctorate in Food, Nutrition and Metabolism from the State University of Campinas (UNICAMP) and Post-Doctorate in Women's Health from the Oswaldo Cruz Foundation (FIOCRUZ) / Fernandes Figueira Institute (IFF).

C. A. B. de Maria is Graduated in Nutrition from the Federal University of Rio de Janeiro (1987), Master in Biochemistry from the Federal University of Rio de Janeiro (1991) and Doctor in Biochemistry from the Federal University of Rio de Janeiro (1995). He is currently a professor at the Department of Public Health at the Biomedical Institute of the Federal University of the State of Rio de Janeiro (UNIRIO). Has experience in Food Science and Nutrition, with emphasis on Chemistry, Physics, Physical Chemistry and Biochemistry of Alim. and of Food Raw Materials.